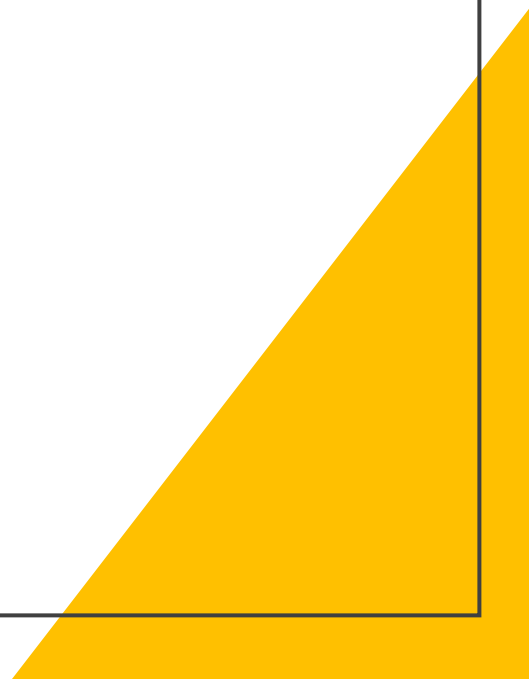


Introduction to Computational Chemistry

Liviu Ungur



Historical Background

- Classical Physics
- The discovery of the electron
- Blackbody radiation
- Photoelectric effect
- Heat capacities
- The Bohr model
- Compton scattering
- Wave-particle duality
- The Heisenberg uncertainty principle



experiments which could be explained only
by considering that **energy is quantized**
(**i.e. energy is not a continuous quantity**)

Wave-Particle duality



Louis de Broglie(1892-1987)
Received Nobel Prize in 1929
"for the idea of wave-like
behavior of matter"

- Scientists have always had trouble describing the nature of light – is it a wave or is it a particle?
- In 1924, de Broglie reasoned that if light can have both wave and particle properties, why not matter?
- From Einstein's relativistic energy equation, the energy of a particle with zero rest mass is the product of its momentum and the speed of light,
 - $E = pc = mc^2$
- Coupling this equation with the energy of a photon, $E = h\nu$, reveals the de Broglie wavelength relationship,
 - $\lambda = \frac{h}{p}$
 - which de Broglie asserted true for both, particles and photons

Wave-Particle duality

- The consequences of this relationship are far reaching (they led to Schrödinger to his theory of wave mechanics), but its basis in nature is unquestionable
 - The diffraction of electrons was observed by Davisson and Germer (1925) and by Thompson (1927)
- The wavelike property of electrons is used in electron microscopes
 - The wavelengths of the electrons can be controlled through an applied voltage, and the small de Broglie wavelengths offer a far more precise probe than an ordinary light microscope
 - In addition, in contrast to electromagnetic radiation of similar wavelengths, the electron beam can be readily focused using electric and magnetic fields

Wave-Particle duality

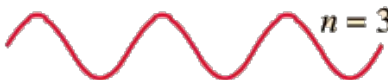
- The de Broglie waves offer a simple, physical argument for the quantized Bohr orbits

For a hydrogen atom:

Electron wave resonance

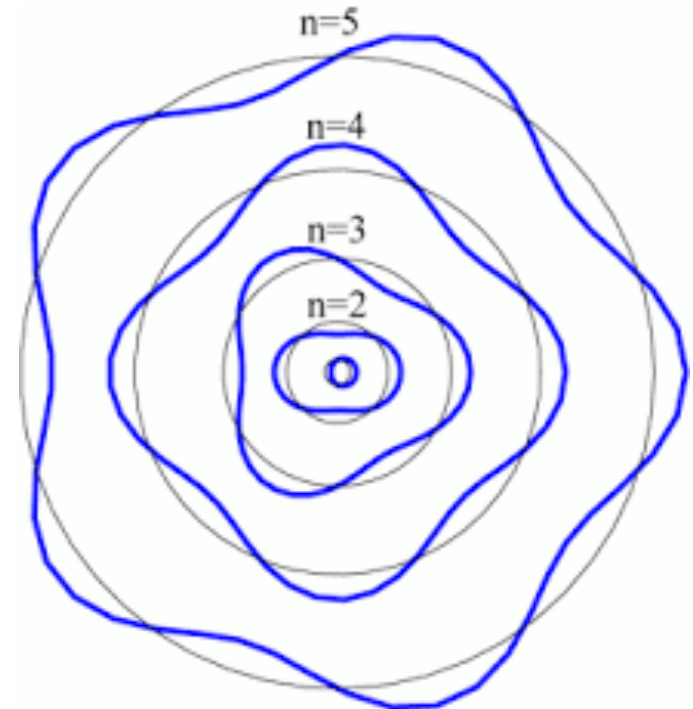
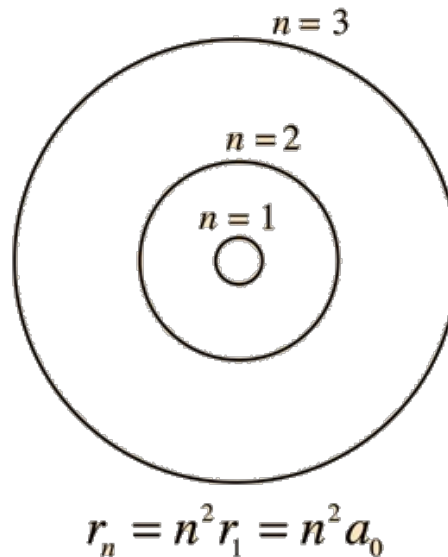
$n=1$
 $\lambda_1 = 2\pi r_1 = 6.28a_0$

$n=2$
 $2\lambda_2 = 2\pi r_2$
 $\lambda_2 = 12.57a_0$

$n=3$
 $3\lambda_3 = 2\pi r_3$
 $\lambda_3 = 18.85a_0$

Wavelengths for hydrogen states.

$a_0 = 0.0529\text{nm} = \text{Bohr radius}$



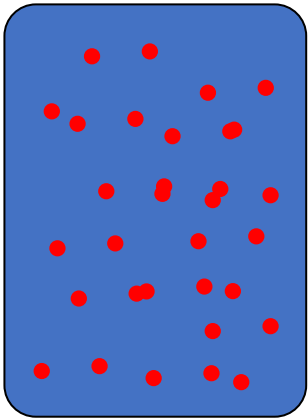
When should be quantum mechanics considered?

- Classical mechanics and quantum mechanics are not competing with each other to describe the world around us
- Classical mechanics can be applied when energy difference between states can be ignored so that the spectrum is regarded as “continuous”
- Quantum description of a system is necessary when the energy spectrum is “discrete”

When should be quantum mechanics considered?

- Boltzmann distribution:

$$\frac{n_i}{n_j} = \frac{g_i}{g_j} e^{-\frac{e_i - e_j}{k_B T}}$$



n_i, n_j – number of gas atoms with energies e_i and e_j
 g_i, g_j – **degeneracy** of each energy state (e_i and e_j)

degeneracy => the number of ways an atom can have an energy within an interval: $e_i - \Delta e < e < e_i + \Delta e$

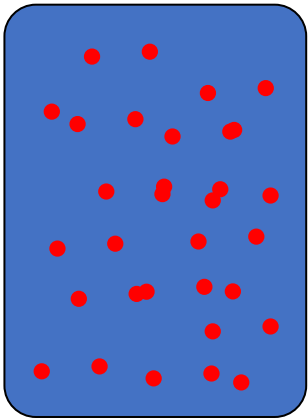
Higher energies = more possibilities to distribute the energy in the system = higher degeneracy (= higher entropy)

$$e_i = \frac{1}{2} m v_i^2 = \frac{1}{2} m (v_{ix}^2 + v_{iy}^2 + v_{iz}^2)$$

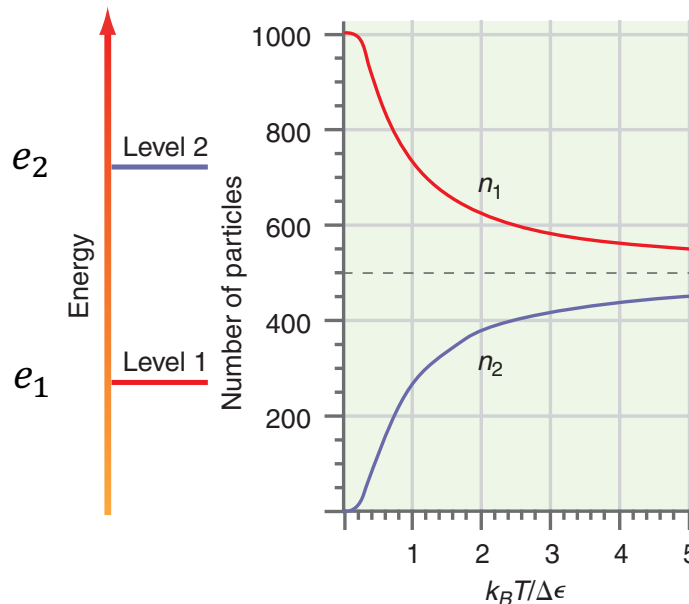
When should be quantum mechanics considered?

- Key concept: Boltzmann distribution:

$$\frac{n_i}{n_j} = \frac{g_i}{g_j} e^{-\frac{e_i - e_j}{k_B T}}$$



Consider a system with 1000 atoms, which can have only two allowed energy states: e_1 and e_2 . Population of each state is:



$$n_1 = \frac{1000}{1 + e^{-\frac{\Delta e}{k_B T}}}$$

$$n_2 = \frac{1000 e^{-\frac{\Delta e}{k_B T}}}{1 + e^{-\frac{\Delta e}{k_B T}}}$$

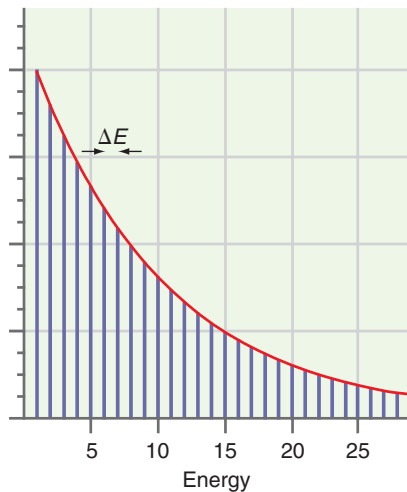
$$e_i = \frac{1}{2} m v_i^2 = \frac{1}{2} m (v_{ix}^2 + v_{iy}^2 + v_{iz}^2)$$

When should be quantum mechanics considered?

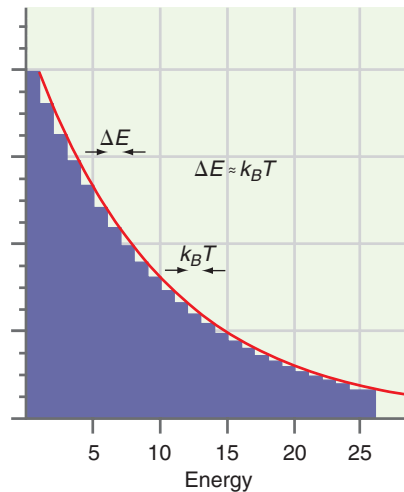
- Key concept: Boltzmann distribution:

$$\frac{n_i}{n_j} = \frac{g_i}{g_j} e^{-\frac{e_i - e_j}{k_B T}} = \frac{g_i}{g_j} e^{-\frac{\Delta E}{k_B T}}$$

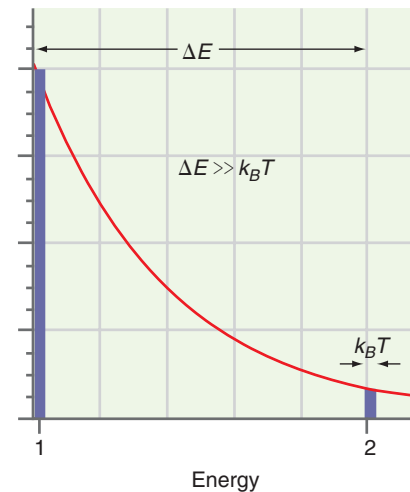
Consider a quantum mechanical system with discrete energy levels



(a)



(b)



(c)

- If $\Delta E < k_B T$ or $\Delta E \approx k_B T$ $\Rightarrow$ spectrum *appear to be continuous* $\Rightarrow$ classical description is enough
- If $\Delta E > k_B T$ $\Rightarrow$ discrete spectrum $\Rightarrow$ quantum description is needed

Wave mechanics

The de Broglie relation enables matter to be described in terms of wave.

Momentum : $mv = \frac{h}{\lambda}; \quad \lambda = \frac{h}{mv}$

Wave properties are far more significant for small objects.

λ (80 kg object moving at 5 ms⁻¹) =

λ (electron moving at 5 ms⁻¹) =

Most molecular properties are derived from **wave mechanics**.

$$\frac{\partial^2 \Psi(x,t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \Psi(x,t)}{\partial t^2} \quad ; \quad \leq \text{Classical nondispersive wave equation in 1D}$$

$\Psi(x, t) = >$ wave function (as example = $A \sin(kx - \omega t)$);

$v = \lambda \nu$ = velocity; $k = 2\pi/\lambda = \omega/v$ = wave vector;

ω = angular frequency;

Why change to wave ? For what we are doing, the mathematics is simpler.

Wave mechanics

$$\frac{\partial^2 \Psi(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \Psi(x, t)}{\partial t^2}$$

Exercise: Show that the wave function: $\Psi(x, t) = A \sin(kx - \omega t + \phi)$ is a solution of the nondispersive wave equation above.

Solution:

$$\text{LHS: } \frac{\partial^2 \Psi(x, t)}{\partial x^2} = \frac{\partial^2 A \sin(kx - \omega t + \phi)}{\partial x^2} = -Ak^2 \sin(kx - \omega t + \phi)$$

$$\text{RHS: } \frac{1}{v^2} \frac{\partial^2 A \sin(kx - \omega t + \phi)}{\partial t^2} = -\frac{\omega^2}{v^2} A \sin(kx - \omega t + \phi)$$

Substitute the relation $k = \omega/v$ in the above expression to obtain the equality.

Wave mechanics

$$\frac{\partial^2 \Psi(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \Psi(x, t)}{\partial t^2}$$

Standing wave: $\Psi(x, t) = \psi(x) \cos \omega t$ (= equivalent of stationary states)

Substitute it into the wave equation:

$$\frac{\partial^2 \Psi(x, t)}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 \Psi(x, t)}{\partial t^2} = 0 \quad \Rightarrow \quad \frac{\partial^2 \psi(x) \cos \omega t}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 \psi(x) \cos \omega t}{\partial t^2} = 0$$

$$\Rightarrow \cos \omega t \frac{\partial^2 \psi(x)}{\partial x^2} - \frac{\psi(x)}{v^2} \frac{\partial^2 \cos \omega t}{\partial t^2} = \cos \omega t \left(\frac{\partial^2 \psi(x)}{\partial x^2} + \frac{\omega^2 \psi(x)}{v^2} \right) = 0$$

$$\frac{\partial^2 \psi(x)}{\partial x^2} + \frac{\omega^2 \psi(x)}{v^2} = 0$$

Substituting $\omega = 2\pi\nu$ and $v = \lambda\nu$ into the above equation we have:

$$\frac{\partial^2}{\partial x^2} \psi(x) + \frac{4\pi^2}{\lambda^2} \psi(x) = 0$$

Schrödinger Equation

The wave equation obtained so far is still “classic”:

$$\frac{\partial^2}{\partial x^2} \psi(x) + \frac{4\pi^2}{\lambda^2} \psi(x) = 0$$

To transform it into “quantum” form we proceed as follows.

The classical expression for the total energy is:

$$E = T + V(x) = \frac{mv^2}{2} + V(x) = \frac{p^2}{2m} + V(x) \quad \Rightarrow \quad p^2 = 2m(E - V(x))$$

Substitute it into the de Broglie relation: $\lambda = \frac{h}{p}$ and insert λ into the above wave equation:

$$\frac{\partial^2}{\partial x^2} \psi(x) + \frac{8\pi^2m}{h^2} [E - V(x)] \psi(x) = 0$$

By substituting the reduced Plank’s constant: $\hbar = h/2\pi$ and reshuffle a bit, we get:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x) \psi(x) = E \psi(x)$$

This is the time independent Schrödinger Equation used to study stationary states of quantum mechanical systems, e.g. 1s and 2pz orbitals of H atom are stationary states.

Schrödinger Equation

The time-independent Schrödinger Equation:

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi(x) + V(x)\psi(x) = E\psi(x)$$

The time-dependent Schrodinger Equation (e.g. when $V(x, t)$):

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \Psi(x, t) + V(x, t)\Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

For stationary states, both equations are satisfied. Therefore:

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = E\Psi(x, t)$$

Also, for stationary states, the $\Psi(x, t) = \psi(x)f(t)$. The wave function takes the form:

$$\Psi(x, t) = \psi(x)e^{-i\left(\frac{E}{\hbar}\right)t}$$

Note that the product: $\Psi^*(x, t)\Psi(x, t) = \psi^*(x)e^{+i\left(\frac{E}{\hbar}\right)t}\psi(x)e^{-i\left(\frac{E}{\hbar}\right)t} = \psi^*\psi$
Is independent on the time-phase.

Operators, Eigenvalues and Eigenfunctions

Schrödinger Equation in 1D can be written in the form:

$$\left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x) = E \psi(x)$$

Or:

$$\hat{H} \psi_n(x) = E_n \psi_n(x)$$

$\hat{H}$ = total energy **operator** (Hamiltonian).

E_n = n^{th} **eigenvalue of the operator $\hat{H}$**

$\psi_n(x)$ = n^{th} **eigenfunction of the operator $\hat{H}$**

This is an example of “eigenvalue equation”:

$$\text{Operator} \times \psi_n(x) = \text{Eigenvalue} \times \psi_n(x)$$

Eigenvalue is a *scalar, real* number.

Operators examples: $\frac{\partial}{\partial x}$, x , Exp , Log , etc.

Operators, Eigenvalues and Eigenfunctions

- Physical observables (each observable corresponds to a Hermitian operator):

- Coordinate $\hat{r}$
- Momentum $\hat{p}$
- Angular momentum $\hat{L} = \hat{r} \times \hat{p}$
- Potential energy $\hat{V}$
- Kinetic energy $\hat{K}$
- Total energy (Hamiltonian) $\hat{H}$
- Electric and magnetic dipole moment $\hat{d}$, $\hat{\mu}$

Representation space:

- Coordinate space $r = (x, y, z)$
- Momentum space $p = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$
- r and p in their representation spaces are continuous functions

Operators, Eigenvalues and Eigenfunctions

Exercise: Verify if the function $\psi(x) = Ae^{ikx} + Be^{-ikx}$ is an eigenfunction of the following operators: $\frac{\partial}{\partial x}$ and $\frac{\partial^2}{\partial x^2}$. Find the eigenvalues, if $\psi(x)$ is an eigenfunction.

$$\frac{\partial}{\partial x}(Ae^{ikx} + Be^{-ikx}) = ikAe^{ikx} - ikBe^{-ikx} = ik(Ae^{ikx} - Be^{-ikx})$$

$\psi(x)$ is not an eigenfunction of $\frac{\partial}{\partial x}$ operator

$$\begin{aligned}\frac{\partial^2}{\partial x^2}(Ae^{ikx} + Be^{-ikx}) &= \frac{\partial}{\partial x}(ik(Ae^{ikx} - Be^{-ikx})) = (ik)^2 Ae^{ikx} + (-ik)^2 Be^{-ikx} \\ &= -k^2(Ae^{ikx} + Be^{-ikx})\end{aligned}$$

$\psi(x)$ - is an eigenfunction of the operator $\frac{\partial^2}{\partial x^2}$ with eigenvalue “ $-k^2$ ”

Orthogonality and Orthonormality

- **Eigenfunctions of a quantum-mechanical operator are orthogonal to each other.**

$$\int \psi_i^*(x) \psi_j(x) dx = 0$$

Orthogonality ensures that all eigenstates are “fully different” from each other

- **All eigenfunctions of a quantum-mechanical operator are normalized**

$$\int \psi_i^*(x) \psi_i(x) dx = 1$$

Normalization ensures that all eigenfunctions have the same “amplitude”, i.e. the probability of finding an electron in a given orbital is 1 (i.e. 100% sure) if the integration is done over entire space.

Orthogonality and Orthonormality

Exercise: Normalize the function $\psi(x) = a(a + x)$ over the interval $0 \leq x \leq a$

$$\begin{aligned} N^2 \int (a(a + x)) (a(a + x)) dx &= N^2 \int (a^2 + ax)(a^2 + ax) dx \\ &= N^2 a^2 \int (a^2 + 2ax + x^2) dx = 1 \end{aligned}$$

Evaluation of the definite integral gives:

$$N^2 a^2 \int_0^a (a^2 + 2ax + x^2) dx = N^2 a^2 \left(a^2 x + ax^2 + \frac{x^3}{3} \right) = N^2 \frac{7a^5}{3}$$

$$N^2 \frac{7a^5}{3} = 1 \quad ; \implies N = \sqrt{\frac{3}{7a^5}}$$

Completeness

A quantum-mechanical operator may have many eigenfunctions.

ALL eigenfunctions of a quantum-mechanical operator form a **COMPLETE** set:

$$\psi(x) = \sum_{n=1}^{\infty} b_n \psi_n(x); \quad \Rightarrow \text{Linear combination}$$

$\psi_n(x)$ = are also called **basis functions**

$\psi(x)$ = a general function constructed by mixing many basis functions

b_n = mixing coefficient of the $\psi_n(x)$ **basis function** into the general wave function

By analogy to real Cartesian tri-dimensional space: any tridimensional vector can be expressed as a linear combination of the three orthogonal (perpendicular) vectors: x, y and z. The three unit vectors **x, y, z** form a **complete set**.

The complete set of eigenfunctions of a given operator could be infinite, but it is not a necessity.

Properties of Operators

- Linearity:
 - 1) $a\hat{O}\psi = \hat{O}a\psi$ $\Rightarrow$ linear operators
 - 2) $\log a\psi \neq a \log \psi$ $\Rightarrow$ non-linear operator
- Commutation:
 - Exercise: Is $\hat{A}\hat{B}\psi = \hat{B}\hat{A}\psi$ for any set of operators $\hat{A}$ and $\hat{B}$? Verify this relation for the position ($\hat{A} = x$) and momenta ($\hat{B} = -i\hbar \frac{\partial}{\partial x} = \hat{p}_x$) operators.
 - $\hat{A}\hat{B}\psi = -i\hbar x \frac{\partial}{\partial x} \psi = -i\hbar x \frac{\partial \psi}{\partial x}$
 - $\hat{B}\hat{A}\psi = -i\hbar \frac{\partial}{\partial x} (x\psi) = -i\hbar \frac{\partial \psi}{\partial x} - i\hbar x \frac{\partial \psi}{\partial x}$
 - $(\hat{A}\hat{B} - \hat{B}\hat{A})\psi = -i\hbar x \frac{\partial \psi}{\partial x} - \left(-i\hbar \frac{\partial \psi}{\partial x} - i\hbar x \frac{\partial \psi}{\partial x}\right) = i\hbar \frac{\partial \psi}{\partial x}$
 - $\Rightarrow$ Operators x and “ $-i\hbar \frac{\partial}{\partial x}$ ” do not commute. This means that **the order of the application of these operators on function ψ matters**

Properties of Operators

- Hermiticity:
- An operator $\hat{O}$ is said to be Hermitian if the following relation is satisfied:
- $$\int \psi_i^*(x) \hat{O} \psi_j(x) dx = \left\{ \int \psi_j^*(x) \hat{O} \psi_i(x) dx \right\}^*$$
- All eigenvalues of a Hermitian Operator are real. (as exercise => prove it).
- All quantum-mechanical operators which correspond to physically observable quantities are Hermitian, and have real-valued eigenvalues.
- As example, Schrodinger Equation has energies as eigenvalues. Hamiltonian matrix is Hermitian.

Properties of commutators

- Heisenberg's suggestion (as formulated by Born and Jordan) amounts to the demand that position operator (x) and momenta operator (p) satisfy the commutation relation
- $[\hat{x}, \hat{p}] = (\hat{x}\hat{p} - \hat{p}\hat{x}) = i\hbar = i \frac{h}{2\pi}$
- The commutation is one of the most important relation in quantum mechanics
- What does all this mean?

Properties of commutators

- First, it implies that Newton's laws are wrong for the description of microscopic particles and must only be an approximation to the true equation of motion
- Classical physics works when making the approximation $\hbar = 0$ has negligible impact (effect) on the evaluated property
- The implication of the commutation relation is more than a minor modification of the form of the observable

Matrix mechanics

- Mathematically there are two possible interpretations:
 - The observable x and p should be regarded as operators
 - The observables x and p should be regarded as matrices

$$\begin{pmatrix} \int \psi_1^*(x) \hat{O} \psi_1(x) dx & \int \psi_1^*(x) \hat{O} \psi_2(x) dx & \cdots & \int \psi_1^*(x) \hat{O} \psi_N(x) dx \\ \int \psi_2^*(x) \hat{O} \psi_1(x) dx & \int \psi_2^*(x) \hat{O} \psi_2(x) dx & \cdots & \int \psi_2^*(x) \hat{O} \psi_N(x) dx \\ \vdots & \vdots & \ddots & \vdots \\ \int \psi_N^*(x) \hat{O} \psi_1(x) dx & \int \psi_N^*(x) \hat{O} \psi_2(x) dx & \cdots & \int \psi_N^*(x) \hat{O} \psi_N(x) dx \end{pmatrix}$$

- The order in which matrices A and B are multiplied together affects the outcome. AB is not necessarily equal to BA .
- The identification of x and p as matrices is the origin of the name “*matrix mechanics*”
- Exercise: Test if the following two matrices commute with each other:
- $A = \begin{pmatrix} -1 & 0 \\ 3 & 2 \end{pmatrix}$; and $B = \begin{pmatrix} 0 & 3 \\ 5 & 1 \end{pmatrix}$

Commutators and anti-commutators:

- If: $(\hat{A}\hat{B} - \hat{B}\hat{A}) = 0 \Rightarrow$ Operators $\hat{A}$ and $\hat{B}$ commute with each other
- If: $(\hat{A}\hat{B} - \hat{B}\hat{A}) \neq 0 \Rightarrow$ Operators $\hat{A}$ and $\hat{B}$ do not commute with each other
- *Shorthand notation:*
 - Commutator: $[\hat{A}, \hat{B}] = (\hat{A}\hat{B} - \hat{B}\hat{A})$
 - Anti-commutator: $[\hat{A}, \hat{B}]_+ = (\hat{A}\hat{B} + \hat{B}\hat{A})$
- If: $[\hat{A}, \hat{B}] = 0$ then all eigenfunctions of the operator $\hat{A}$ are also eigenfunctions of the operator $\hat{B}$, and vice-versa

Commutator algebra

- $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$
- $[\hat{A}, \hat{A}^n] = 0$, for $n = 1, 2, 3, \dots$
- $[\alpha \hat{A}, \hat{B}] = [\hat{A}, \alpha \hat{B}] = \alpha [\hat{A}, \hat{B}]$, where $\alpha = \text{constant}$
- $[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]$
- $[\hat{A} + \hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}]$
- $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$
- $[\hat{A}\hat{B}, \hat{C}] = [\hat{A}, \hat{C}]\hat{B} + \hat{A}[\hat{B}, \hat{C}]$
- $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$
- Exercise: verify the above commutator relations

Vectors

- Before we represent the operators in a specific space, we first review simple vector algebra operations:
- Assume that $\vec{A} = (A_x, A_y, A_z)$ and $\vec{B} = (B_x, B_y, B_z)$
- Dot product:
- $\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$
- Cross Product (= vector product):
- $\vec{A} \times \vec{B} = (A_y B_z - B_y A_z, A_z B_x - A_x B_z, A_x B_y - A_y B_x) = \begin{vmatrix} x & y & z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

Operators represented in Coordinate Space

- Coordinate: $\hat{r} \rightarrow \vec{r} = (x, y, z)$
- Momentum: $\hat{p} \rightarrow (p_x, p_y, p_z) = -i\hbar\vec{\nabla} = -i\hbar\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$
- Angular momentum: $\hat{L} = \hat{r} \times \hat{p} \rightarrow (yp_z - zp_y, zp_x - xp_z, xp_y - yp_x)$
- Potential energy: $\hat{V} \rightarrow V(x, y, z)$

- Kinetic Energy: $\hat{K} = \frac{1}{2m}\vec{p}^2 \rightarrow \frac{1}{2m}(-i\hbar\vec{\nabla})^2 = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial z^2}\right)$
- Hamiltonian (total energy): $\hat{H} = \hat{K} + \hat{V} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}, \frac{\partial^2}{\partial y^2}, \frac{\partial^2}{\partial z^2}\right) + V(x, y, z)$

Operators represented in Coordinate Space

- The Kinetic energy operator is independent of time (in contrast to classical physics, where $\hat{K} = \frac{\vec{p}^2}{2m} = \frac{mv^2}{2} = \frac{m}{2} \left(\frac{\partial \vec{r}}{\partial t} \right)^2$)
- For molecular and material systems not under the influence of an external field, the potential energy $V(x, y, z)$ is independent of time. The time-dependent Schrodinger equation can be reduced to time-independent Schrodinger equation in this case
- In coordinate space the Schrodinger equation is:

$$\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z) \right] \psi(x, y, z) = E\psi(x, y, z)$$

Operators represented in Coordinate Space

- $\left[-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(x, y, z) \right] \psi(x, y, z) = E\psi(x, y, z)$
- Verify the following:
 - $[x, p^2] = \dots = 2\hbar^2 \frac{\partial}{\partial x}$
 - $[x, H] = [x, K + V] = [x, K] + [x, V] = \dots = \frac{\hbar^2}{m} \frac{\partial}{\partial x} = \frac{i\hbar}{m} p$
 - $[p, H] = \dots = -i\hbar \frac{\partial V}{\partial x}$
- For 3-dimensional systems, we note $[x, p_y] = [x, p_z] = 0$. The rest of the operation is similar to the 1-D systems

Born Interpretation

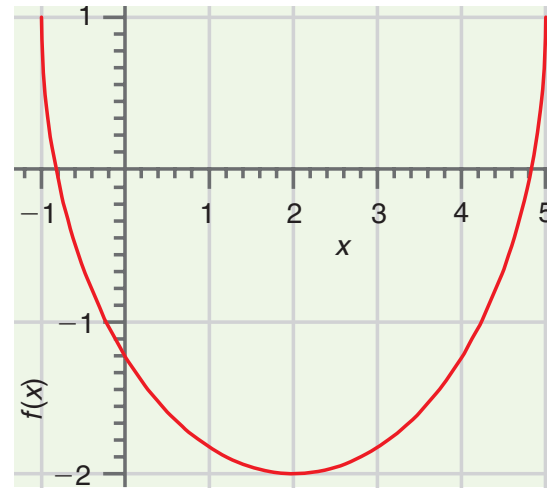
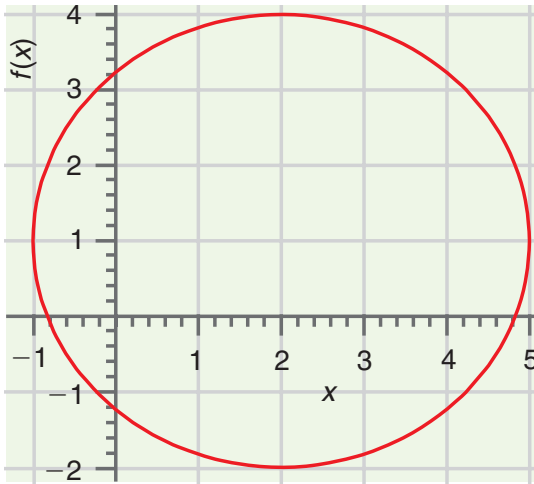
- In 1926, Born suggested that the product $\Psi^*\Psi$ should be regarded as a “probability density”
- Ψ can be a complex function, since it is the product with its complex conjugate which specifies the real physical probability of finding a particle in a particular state
- It gives the probability when it is integrated over all space which is of interest
- It leads to the normalization condition when integrated over all space
- $\int \Psi^*\Psi d\tau = 1$

The Quantum Mechanical Postulates

- Postulate 1
- The state of a quantum mechanical particle is completely specified by a wave function $\Psi(x, t)$.
- The probability that the particle will be found at time t_0 in a spatial interval of width dx centered at x_0 is given by
- $P(x_0, t_0)dx = \Psi^*(x_0, t_0)\Psi(x_0, t_0)dx = |\Psi(x_0, t_0)|^2 dx$
- $\int_{-\infty}^{+\infty} \Psi^*(x_0, t_0)\Psi(x_0, t_0)dx = 1$

The Quantum Mechanical Postulates

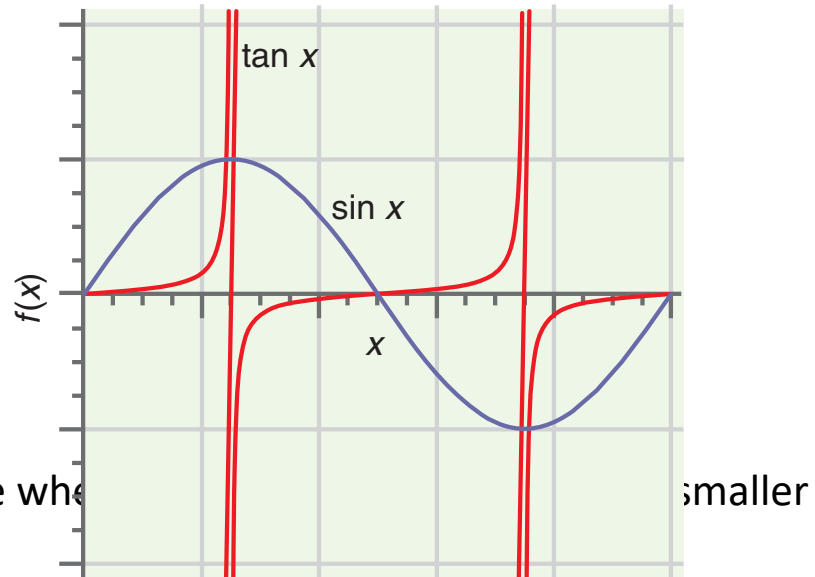
- Requirements for Ψ
- Ψ must be single-valued function



- Ψ has two values for almost each value of x Ψ has only one value for each value of x
- Ψ is not good wave function Ψ is good wave function

The Quantum Mechanical Postulates

- Requirements for Ψ
- Ψ must be continuous function



- $f(x)$ exists
- $\lim_{x \rightarrow a} f(x)$ exists and has the same value whether x approaches a from larger or smaller values
- $\lim_{x \rightarrow a} f(x) = f(a)$
- The wave function *cannot have an infinite value* within a given interval.

•

The Quantum Mechanical Postulates

- Postulate 2
- For every measurable property of a system such as: position, momentum and energy, there exists a corresponding operator in quantum mechanics.
- An experiment in the laboratory to measure a value for such an observable is simulated in the theory by operating on the wave function of the system with the corresponding operator.

The Quantum Mechanical Postulates

- Postulate 3
- In any single measurement of the observable that corresponds to operator $\hat{A}$, the only values that will ever be measured are the *eigenvalues* of that operator.
- $\hat{H}\Psi_n(x, t) = E_n\Psi_n(x, t)$

The Quantum Mechanical Postulates

- Postulate 4
- If a system is in a state described by the wave function $\Psi(x, t)$ and the value of the observable a is measured once on each of many identically prepared systems, the *average value* (also *expectation value*) of all of these measurements is given by
- $$\langle a \rangle = \frac{\int_{-\infty}^{+\infty} \Psi^*(x, t) \hat{A} \Psi(x, t) dx}{\int_{-\infty}^{+\infty} \Psi^*(x, t) \Psi(x, t) dx}$$
- If the $\Psi(x, t)$ is normalized, the denominator $\int_{-\infty}^{+\infty} \Psi^*(x, t) \Psi(x, t) dx = 1$

The Quantum Mechanical Postulates

- Postulate 4
- Case 1: $\Psi(x, t)$ is a normalized eigenstate of the operator $\hat{A}$, e.g. $\varphi_i(x, t)$

$$\hat{A}\varphi_i(x, t) = a_i\varphi_i(x, t)$$

- Expectation value is:

$$\langle a \rangle = \frac{\int_{-\infty}^{+\infty} \varphi_i^*(x, t) \hat{A}\varphi_i(x, t) dx}{\int_{-\infty}^{+\infty} \varphi_i^*(x, t) \varphi_i(x, t) dx} = \frac{a_i \int_{-\infty}^{+\infty} \varphi_i^*(x, t) \varphi_i(x, t) dx}{\int_{-\infty}^{+\infty} \varphi_i^*(x, t) \varphi_i(x, t) dx} = a_i$$

The Quantum Mechanical Postulates

- Postulate 4
- Case 2: $\Psi(x, t)$ is not an eigenstate of the operator $\hat{A}$. Then $\Psi(x, t)$ can be expanded in the complete set of (orthonormal) eigenstates of the operator $\hat{A}$:

$$\Psi(x, t) = \sum_n b_n \varphi_n(x, t)$$

- Expectation value is:

$$\begin{aligned} \langle a \rangle &= \int_{-\infty}^{+\infty} \Psi^*(x, t) \hat{A} \Psi(x, t) dx = \int_{-\infty}^{+\infty} \left[\sum_{m=1}^{\infty} b_m^* \varphi_m^*(x, t) \right] \left[\sum_{n=1}^{\infty} a_n b_n \varphi_n(x, t) \right] dx = \\ &= \sum_{m=1}^{\infty} b_m^* b_n a_n \int_{-\infty}^{+\infty} \varphi_m^*(x, t) \varphi_n(x, t) dx = \sum_{m=1}^{\infty} b_m^* b_m a_m = \sum_{m=1}^{\infty} |b_m|^2 a_m \end{aligned}$$

- b_n = expansion coefficient of eigenstate n into the $\Psi(x, t)$
- $|b_n|^2$ = weight of the eigenstate n into the $\Psi(x, t)$

Superposition principle

Before a measurement is made, the wavefunction of the particle is not known. It is best represented as a combination of all the possible wavefunctions associated with each energy level.

$$\Psi(x, t) = \sum_{\infty} b_n \varphi_n(x, t)$$
$$\langle a \rangle = \sum_{n=1} |b_n|^2 a_n$$

- b_n = expansion coefficient of eigenstate n into the $\Psi(x, t)$
- $|b_n|^2$ = weight of the eigenstate n into the $\Psi(x, t)$
- $\langle a \rangle$ = expectation value is a weighted average
- Individual measurement = find only one eigenvalue (one b_n).
- We need many measurements to find the values of all b_n

Superposition principle



Schrodinger's cat paradox.

Put in a box a cat and a bottle of poisonous gas.
The poisonous gas can be randomly released by a mechanism.
Before opening the box, we do not know if the cat is dead or alive.

“Opening the box” = measurement process in Quantum Mechanics

If we were to repeat this measurement a large number of times,
then in approximately 50% of the cases the cat will be alive.

Schrodinger's Cat

$$\Psi_{\text{cat}} = \Psi_{\text{alive}} + \Psi_{\text{not alive}}$$

Measurement process in QM is “**active**”: during measurement we collapse the wave function into a certain eigenvalue of the operator of the measured property! The initial wave function is thus destroyed (all other information contained in the wave function is lost).

A question by Albert Einstein, about the importance of measurement in QM:
“Do you really think that the moon is not there when we are not looking at it?”

Time evolution

- Postulate 5
- The evolution in time of a quantum mechanical system is governed by time-dependent Schrodinger equation

$$\hat{H}\Psi_n(x, t) = i\hbar \frac{\partial \Psi_n(x, t)}{\partial t}$$

Hydrogen atom - the only quantum system which can be solved exactly (analytically)

Coulomb Interaction between two charges:

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{|r_1 - r_2|}$$

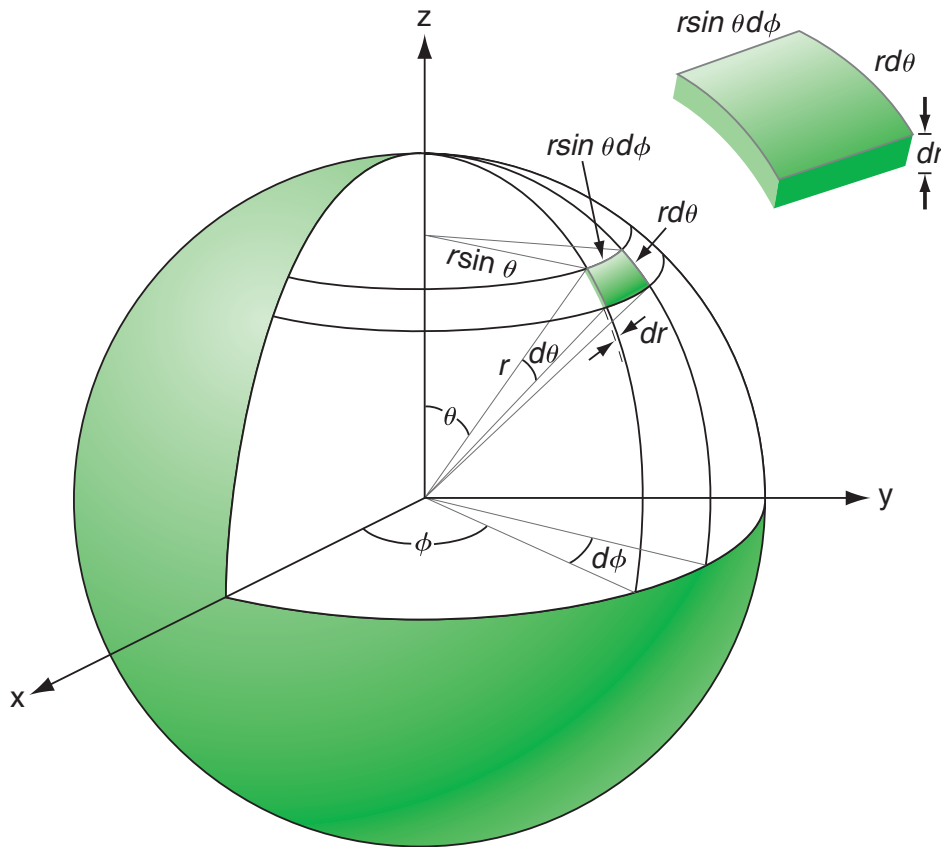
Q_i = magnitude of the charge i

r_i = position in space of charge i

$|r_1 - r_2|$ = the distance between the two charges

ϵ_0 = constant, permittivity of vacuum

Spherical coordinate system



Spherical polar coordinates

$$\begin{aligned}
 x &= r \sin \theta \cos \varphi \\
 y &= r \sin \theta \sin \varphi \\
 z &= r \cos \theta \\
 r^2 &= x^2 + y^2 + z^2 \\
 \cos \theta &= \frac{z}{\sqrt{x^2 + y^2 + z^2}} \\
 \tan \varphi &= \frac{y}{x}
 \end{aligned}$$

Central Force problem

Central force = is the force derived from a spherically symmetric potential energy function.

$$\mathbf{F}(x, y, z) = -\nabla V(x, y, z) = -\frac{\partial V}{\partial x} \mathbf{i} - \frac{\partial V}{\partial y} \mathbf{j} - \frac{\partial V}{\partial z} \mathbf{k}$$

Where $(\mathbf{i}, \mathbf{j}, \mathbf{k})$ are unit vectors along the three Cartesian axes (x, y, z)

$V(x, y, z)$ is a function of $r = \sqrt{x^2 + y^2 + z^2}$ only. $\rightarrow$ pass to spherical coordinate system (r, θ, φ)

Partial derivatives of $V(x, y, z)$ are:

$$\begin{aligned} \left(\frac{\partial V}{\partial x} \right)_{y,z} &= \frac{\partial V}{\partial r} \left(\frac{\partial r}{\partial x} \right)_{y,z} = \frac{x}{r} \frac{\partial V}{\partial r} \\ \left(\frac{\partial V}{\partial y} \right)_{x,z} &= \frac{\partial V}{\partial r} \left(\frac{\partial r}{\partial y} \right)_{x,z} = \frac{y}{r} \frac{\partial V}{\partial r} \\ \left(\frac{\partial V}{\partial z} \right)_{x,y} &= \frac{\partial V}{\partial r} \left(\frac{\partial r}{\partial z} \right)_{x,y} = \frac{z}{r} \frac{\partial V}{\partial r} \end{aligned}$$

Substitute them into the Force equation:

Central Force problem

Substitute them into the Force equation:

$$F(x, y, z) = -\nabla V(x, y, z) = -\frac{x}{r} \frac{\partial V}{\partial r} \mathbf{i} - \frac{y}{r} \frac{\partial V}{\partial r} \mathbf{j} - \frac{z}{r} \frac{\partial V}{\partial r} \mathbf{k} = -\frac{1}{r} \frac{\partial V}{\partial r} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = -\frac{dV(r)}{dr} \frac{\mathbf{r}}{r}$$

$$F(r) = -\frac{dV(r)}{dr} \frac{\mathbf{r}}{r}$$

A central force is radially directed.

The quantum mechanical Hamiltonian for an electron in central force is:

$$\hat{H} = \hat{K} + \hat{V} = -\frac{\hbar^2}{2m_e} \nabla^2 + V(r)$$

Where $\nabla^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$

Central Force problem

In spherical coordinate system (r, θ, φ) , the gradient ∇ is:

$$\begin{aligned}\frac{\partial}{\partial x} &= \sin \theta \cos \varphi \frac{\partial}{\partial r} + \frac{\cos \theta \cos \varphi}{r} \frac{\partial}{\partial \theta} - \frac{\sin \varphi}{r \sin \theta} \frac{\partial}{\partial \varphi} \\ \frac{\partial}{\partial y} &= \sin \theta \sin \varphi \frac{\partial}{\partial r} + \frac{\cos \theta \sin \varphi}{r} \frac{\partial}{\partial \theta} + \frac{\cos \varphi}{r \sin \theta} \frac{\partial}{\partial \varphi} \\ \frac{\partial}{\partial z} &= \cos \theta \frac{\partial}{\partial r} - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta}\end{aligned}$$

In order to derive the Laplacian operator $\nabla^2 = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right)$ in spherical coordinate system we need to square the partial derivatives above and sum them up. We find:

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{1}{r^2} \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}$$

On the other hand, the square of the angular momentum operator, is:

$$L^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right)$$

Central Force problem

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{1}{r^2} \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}$$

On the other hand, the square of the angular momentum operator, is:

$$\hat{L}^2 = -\hbar^2 \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right)$$

Therefore, the Laplacian can be written as:

$$\begin{aligned} \nabla^2 &= \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2} + \frac{1}{r^2} \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{r^2} \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} = \\ &= \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \left(\frac{\partial^2}{\partial \theta^2} + \cot \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \end{aligned}$$

Or:

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{1}{r^2 \hbar^2} \hat{L}^2$$

The Hamiltonian becomes:

$$\hat{H} = -\frac{\hbar^2}{2m_e} \nabla^2 + V(r) = -\frac{\hbar^2}{2m_e} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{1}{r^2 \hbar^2} \hat{L}^2 \right) + V(r) =$$

Central Force problem

The Hamiltonian of one electron in central force becomes:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla^2 + V(r) = -\frac{\hbar^2}{2m_e}\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} - \frac{1}{r^2\hbar^2}\hat{L}^2\right) + V(r)$$

Or:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\left(\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right) + \frac{1}{2m_er^2}\hat{L}^2 + V(r)$$

We notice the following commutator properties:

$$[\hat{H}, \hat{L}^2] = [\hat{K}, \hat{L}^2] + [\hat{V}, \hat{L}^2] = 0, \quad \text{if } V = V(r)$$

Since $\hat{L}_z = -i\hbar\frac{\partial}{\partial\varphi}$, we also notice that:

$$[\hat{H}, \hat{L}_z] = [\hat{K}, \hat{L}_z] + [\hat{V}, \hat{L}_z] = 0, \quad \text{if } V = V(r)$$

Exercise: prove the above two expressions

Central Force problem

As a result of the two commutator relations:

$$[\hat{H}, \hat{L}^2] = [\hat{H}, \hat{L}_z] = 0, \quad \text{if } V = V(r)$$

The operators $\hat{H}$, $\hat{L}^2$, and $\hat{L}_z$ have a common set of eigenstates for a particle in central force! Therefore, the following equations are valid!

$$\hat{H}\Psi = E\Psi$$

$$\hat{L}^2\Psi = l(l+1)\hbar^2\Psi, \quad l = 0, 1, 2, 3, \dots$$

$$\hat{L}_z\Psi = m\hbar\Psi, \quad m = -l, -l+1, -l+2, \dots, l-2, l-1, l$$

The Hamiltonian operator can be re-written:

$$\hat{H} = -\frac{\hbar^2}{2m_e} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) + \frac{\hbar^2}{2m_e} \frac{l(l+1)}{r^2} + V(r)$$

Or:

$$-\frac{\hbar^2}{2m_e} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \right) \Psi + V(r)\Psi = E\Psi$$

Central Force problem

The eigenstates of the $\hat{L}^2$ operator are Spherical Harmonics $Y_l^m(\theta, \varphi)$. Since the operator $\hat{L}^2$ does not involve r , the solution of the Hamiltonian can be written as a product between Spherical Harmonic $Y_l^m(\theta, \varphi)$ and a radial function $R(r)$:

$$\Psi = \mathbf{R}(\mathbf{r})\mathbf{Y}_l^m(\boldsymbol{\theta}, \boldsymbol{\varphi}) = R(r) \sqrt{\frac{2l+1}{4\pi} \frac{(l-|m|)!}{(l+|m|)!}} P_l^{|m|}(\cos \theta) e^{im\varphi}$$

Insert this form of Ψ in the Hamiltonian operator, :

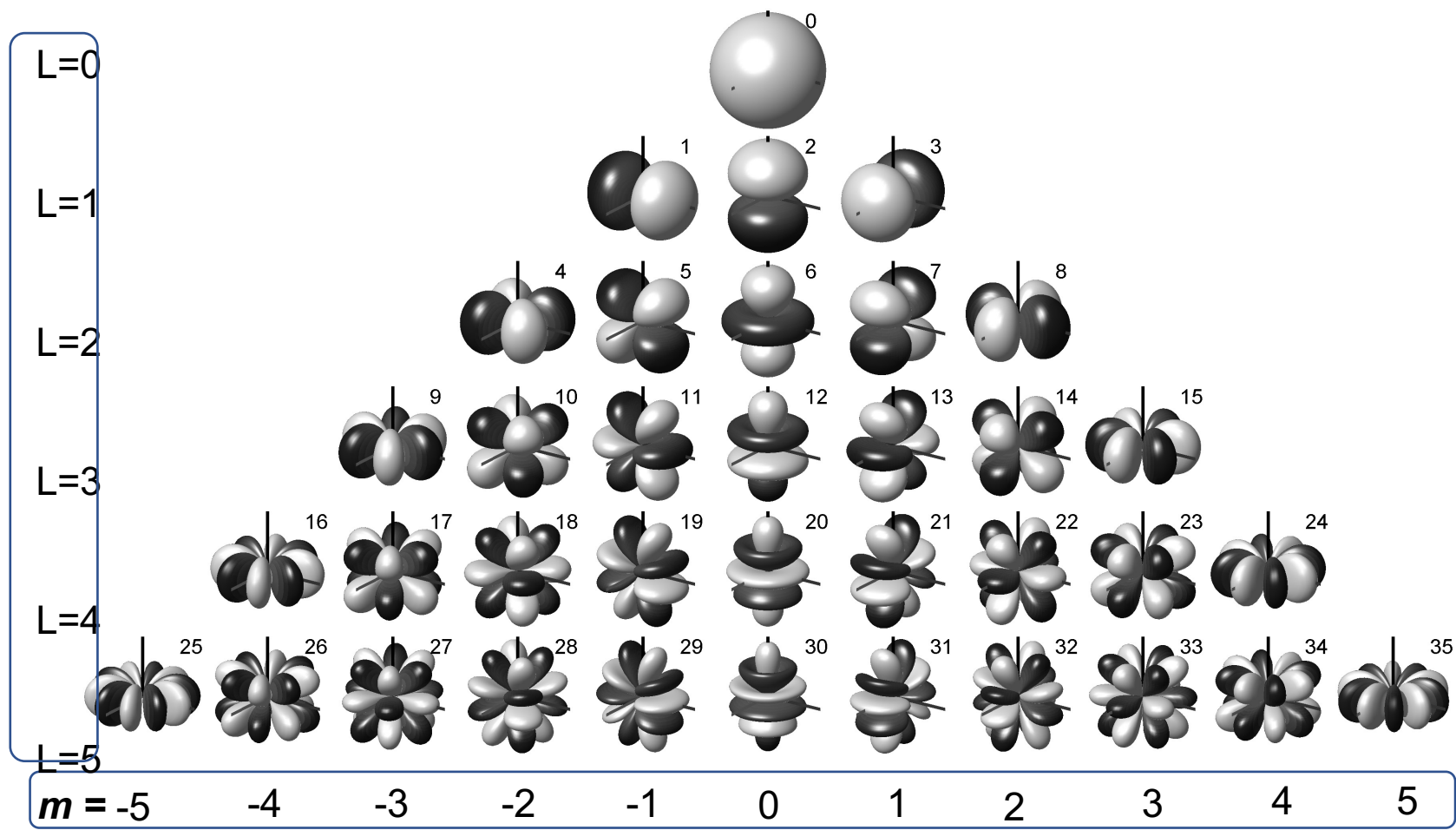
$$-\frac{\hbar^2}{2m_e} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \right) R(r) Y_l^m(\theta, \varphi) + V(r) R(r) Y_l^m(\theta, \varphi) = E R(r) Y_l^m(\theta, \varphi)$$

and divide both sides by $Y_l^m(\theta, \varphi)$:

$$\boxed{-\frac{\hbar^2}{2m_e} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} \right) R(r) + V(r) R(r) = E R(r)}$$

For any one-particle problem in spherically symmetric potential $V(r)$, the stationary wave functions are $R(r)Y_l^m(\theta, \varphi)$, where the radial function $R(r)$ satisfies the above differential equation

Shape of the (real-valued) *Spherical Harmonics*:



Radial Wave Function

The **radial wave function** can only be solved for 1-electron systems.

For 2- (and more) -electronic systems, the radial equation cannot be solved analytically. Numerical methods allow for (almost) any accuracy needed.

The **radial wave function** is tabulated inside the **basis sets information**.

FIGURE 6.6 Energy levels of the hydrogen atom.

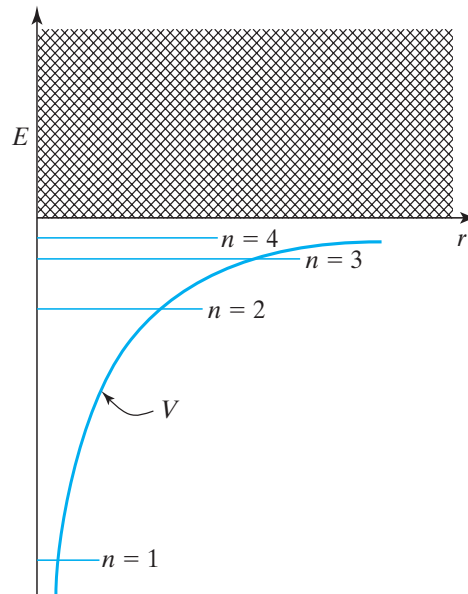


TABLE 6.1 Radial Factors in the Hydrogenlike-Atom Wave Functions

$$R_{1s} = 2\left(\frac{Z}{a}\right)^{3/2} e^{-Zr/a}$$

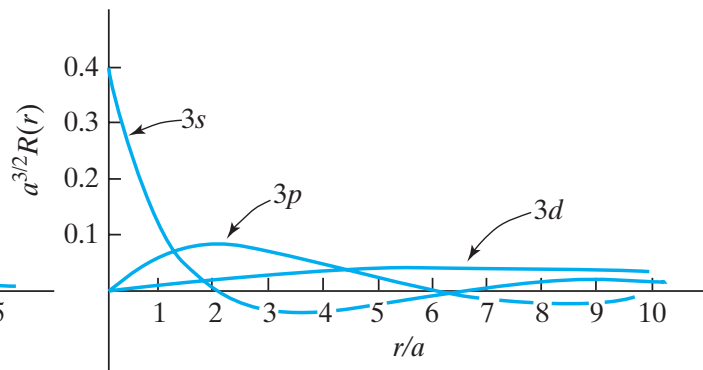
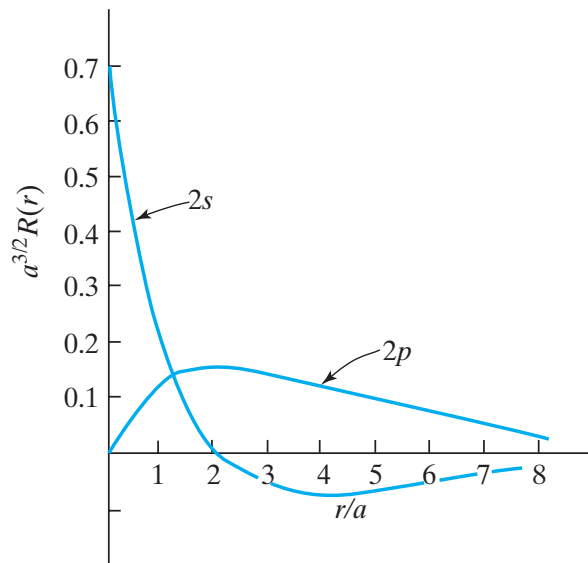
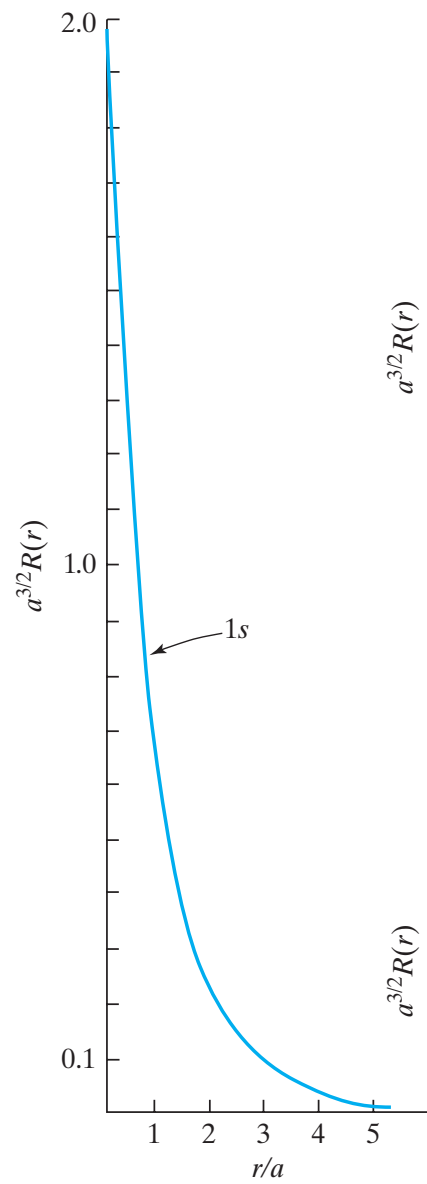
$$R_{2s} = \frac{1}{\sqrt{2}}\left(\frac{Z}{a}\right)^{3/2} \left(1 - \frac{Zr}{2a}\right) e^{-Zr/2a}$$

$$R_{2p} = \frac{1}{2\sqrt{6}}\left(\frac{Z}{a}\right)^{5/2} r e^{-Zr/2a}$$

$$R_{3s} = \frac{2}{3\sqrt{3}}\left(\frac{Z}{a}\right)^{3/2} \left(1 - \frac{2Zr}{3a} + \frac{2Z^2 r^2}{27a^2}\right) e^{-Zr/3a}$$

$$R_{3p} = \frac{8}{27\sqrt{6}}\left(\frac{Z}{a}\right)^{3/2} \left(\frac{Zr}{a} - \frac{Z^2 r^2}{6a^2}\right) e^{-Zr/3a}$$

$$R_{3d} = \frac{4}{81\sqrt{30}}\left(\frac{Z}{a}\right)^{7/2} r^2 e^{-Zr/3a}$$

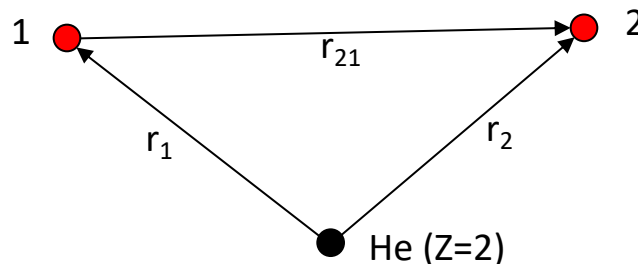


Helium atom

- It is not possible to obtain exact wave functions for any system with more than one electron, however, we can obtain numerical solutions as accurate as we please
- The Helium atom is the first and simplest many electron atom. The Hamiltonian operator for He atom must include electron-electron repulsion term:

$$\hat{H} = \left(-\frac{1}{2}\nabla_1^2 - \frac{2}{r_1}\right) + \left(-\frac{1}{2}\nabla_2^2 - \frac{2}{r_2}\right) + \frac{1}{r_{12}}$$

Where r_{12} is the distance between the two electrons: $r_{12} = r_2 - r_1$



Helium atom

- Note that we have labelled the electrons 1 and 2. Can we really distinguish each of them?
- The $\frac{1}{r_{12}}$ term represents the Coulomb repulsion between electrons (electrons wish to avoid each other as much as possible)
- This means that their individual angular momenta will fluctuate, i.e. they are not quantized. Hence, individual orbital quantum numbers l_1 and l_2 of each electron are not good quantum numbers
- However, the sum of orbital momenta l_1 and l_2 : ($L = l_1 + l_2$) is conserved (i.e. does not change with time, is stable and represents a good quantum number of the entire system).

Total angular momenta

- Total angular momenta for a multi-electronic atoms:

$$L = l_1 + l_2 + l_3 + \dots = \sum_{i=1}^N l_i$$

- Total spin momenta for a multi-electronic atoms:

$$S = s_1 + s_2 + s_3 + \dots = \sum_{i=1}^N s_i$$

- The z-components of L and S are given by:

$$L_z = \sum_{i=1}^N l_{zi} = \sum_{i=1}^N m_i = M$$

$$S_z = \sum_{i=1}^N s_{zi} = \sum_{i=1}^N m_{si} = M_s$$

- Note that the z-components of L and S add as scalars (i.e. as usual numbers)

Total angular momenta

- Thus, the states of many-electron atoms can be expected to have orbital angular momentum quantum numbers:

$$L = 0, 1, 2, 3 \dots$$
$$L_z = M_L = 0, \pm 1, \pm 2, \pm 3, \dots, \pm L$$

- Nevertheless, l_i for each electron is still a useful quantum number because it gives the average orbital angular momentum of electron i
- By ***averaging the electron-electron interaction***, we can get an ***approximate*** radial force Hamiltonian.
- Angular momentum (L, S) in the case of ***free atoms and ions*** is conserved.

Orbital approximation

- The true wave function of a multi-electronic Hamiltonian depends on the coordinates of all electrons. As example, the Hamiltonian for He:

$$\hat{H} = \left(-\frac{1}{2} \nabla_1^2 - \frac{2}{r_1} \right) + \left(-\frac{1}{2} \nabla_2^2 - \frac{2}{r_2} \right) + \frac{1}{r_{12}}$$

has eigenstates:

$$\psi(r_1, \theta_1, \varphi_1, \sigma_1; r_2, \theta_2, \varphi_2, \sigma_2)$$

- Orbital approximation** consists in writing the above wave function as a **product of effective mono-electronic wave functions**:

$$\psi(r_1, \theta_1, \varphi_1, \sigma_1; r_2, \theta_2, \varphi_2, \sigma_2) \sim \varphi_1(r_1, \theta_1, \varphi_1, \sigma_1) \varphi_2(r_2, \theta_2, \varphi_2, \sigma_2)$$

This is an approximation, because the exact wave function may have a much more complex form (i.e. it may include quadratic terms, mixed variable terms, etc.), but if we wish to solve $\hat{H}$ for real molecules, we have no alternative...

Orbital approximation: consequences

$$\hat{H} = \left(-\frac{1}{2} \nabla_1^2 - \frac{2}{r_1} \right) + \left(-\frac{1}{2} \nabla_2^2 - \frac{2}{r_2} \right) + \frac{1}{r_{12}}$$

- The term r_{12} in the above Hamiltonian kills the “spherical symmetry”, since the vector $\mathbf{r}_{12} = \mathbf{r}_1 - \mathbf{r}_2$ does not originate in the origin of coordinates.
- This term makes it impossible to solve the above Hamiltonian exactly → further approximations are needed.
- Mean-Field Approximation (MFA): ***each electron moves in an average electrostatic field created by nuclei + all other electrons.***
- MFA neglects the true electron-electron correlated motion (i.e. ***electron correlation***). In reality – electrons avoid each other as much as possible, since they repel each other.

Orbital approximation

- With this approximation, it means that we can still use the $l_i = 0, 1, 2, 3 \dots$ just as for the hydrogen atoms
- Hence, we arrive at four quantum numbers for each electron: n_i, l_i, m_i, s_{iz}
- This one-electron approximation is known as ***orbital approximation*** and together with the ***Pauli exclusion principle*** explains the Periodic Table

Electron spin

- By analogy with the orbital angular momentum operators, we have spin angular momentum operators $\hat{s}^2$ and $\hat{s}_z$. They quantify the “angular motion of electron around its own axis of rotation”.
- The eigenfunctions of $\hat{s}^2$ and $\hat{s}_z$ for an electron are denoted α and β :

$$\int \alpha^* \alpha d\sigma = \int \beta^* \beta d\sigma = 1$$

$$\int \alpha^* \beta d\sigma = \int \beta^* \alpha d\sigma = 0$$

$$\hat{s}^2 \alpha = s(s+1)\alpha$$

$$\hat{s}^2 \beta = s(s+1)\beta$$

$$\hat{s}_z \alpha = +\frac{1}{2}\alpha; \quad \hat{s}_z \beta = -\frac{1}{2}\beta$$

Electron spin

- Thus, the spin wave function with spin quantum numbers $s = \frac{1}{2}$ has the spin magnetic quantum number: $m_s = +\frac{1}{2}$ or $m_s = -\frac{1}{2}$
- We use lower case letters for spin quantum number of one electron (s and m_s) and capital letters for the total spin quantum numbers for many electron systems (S and M_S).

Pauli Principle

- Pauli Principle states that an electronic wave function $\psi(1,2, \dots)$ is antisymmetric with respect to interchange of any pair of two electrons:

$$\hat{P}_{12}\psi(1,2, \dots) = \psi(2,1, \dots)$$

$$\psi(1,2, \dots) = -\psi(2,1, \dots)$$

- Note that the probability density $\psi^*(1,2, \dots)\psi(1,2, \dots)$ is not changed by the permutation operator, only if the electrons 1 and 2 are identical,

$$\psi^*(1,2, \dots)\psi(1,2, \dots) = \psi^*(2,1, \dots)\psi(2,1, \dots)$$

Because the probability density is a physically measurable quantity (observable)

- The anti-symmetry requirement is satisfied in the orbital approximation by a ***Slater determinant*** form of the wave function.

Helium atom

- Since the two electrons are indistinguishable, we cannot put labels to them. The two wave functions are both equivalent

$$\psi(1,2) = 1s(r_1)\alpha_1 1s(r_2)\beta_2$$

$$\psi(2,1) = 1s(r_2)\alpha_2 1s(r_1)\beta_1$$

- There are two possible linear combinations between them:

$$\Psi_+ = \psi(1,2) + \psi(2,1) = 1s(r_1)\alpha_1 1s(r_2)\beta_2 + 1s(r_2)\alpha_2 1s(r_1)\beta_1$$

$$\Psi_- = \psi(1,2) - \psi(2,1) = 1s(r_1)\alpha_1 1s(r_2)\beta_2 - 1s(r_2)\alpha_2 1s(r_1)\beta_1$$

- The effect of permutation operator on the interchange of two electrons on Ψ_+ and Ψ_- gives:

$$\hat{P}_{12}\Psi_+ = \hat{P}_{12}(\psi(1,2) + \psi(2,1)) = \psi(2,1) + \psi(1,2) = \Psi_+$$

$$\hat{P}_{12}\Psi_- = \hat{P}_{12}(\psi(1,2) - \psi(2,1)) = \psi(2,1) - \psi(1,2) = -\Psi_-$$

Only Ψ_- corresponds to Pauli principle! $L = 0, S = 0, J = 0 \rightarrow {}^1S_0$

$$\Psi({}^1S_0) = \frac{1}{\sqrt{2}}(1s(r_1)1s(r_2))(\alpha_1\beta_2 - \alpha_2\beta_1)$$

Helium atom: excited state $1s2s$

- Let us now consider the first excited state of He, $1s2s$.
- In this case, because the electrons are in different orbitals and are indistinguishable, the spatial eigenfunction must be written down as linear combination of $1s(r_1)2s(r_2)$ and $1s(r_2)2s(r_1)$
- This results in symmetric and anti-symmetric set of spatial wave functions:

$$\begin{aligned} \frac{1}{\sqrt{2}}(1s(r_1)2s(r_2) + 1s(r_2)2s(r_1)) & \quad \text{symmetric} \\ \frac{1}{\sqrt{2}}(1s(r_1)2s(r_2) - 1s(r_2)2s(r_1)) & \quad \text{anti-symmetric} \end{aligned}$$

Helium atom: excited state $1s2s$

- The possible spin eigenfunctions are:

$$\left. \begin{array}{l} \alpha_1\alpha_2 \\ \beta_1\beta_2 \\ \frac{1}{\sqrt{2}}(\alpha_1\beta_2 + \alpha_2\beta_1) \\ \frac{1}{\sqrt{2}}(\alpha_1\beta_2 - \alpha_2\beta_1) \end{array} \right\} \text{symmetric}$$

- In order to build eigenfunction which satisfy the Pauli principle, we must combine a symmetric function with one anti-symmetric.

Helium atom: excited state $1s2s$

- The following possible eigenstates are allowed:

$\frac{1}{\sqrt{2}}(1s(r_1)2s(r_2) - 1s(r_2)2s(r_1))\alpha_1\alpha_2$	$S=1, M_s=+1$	} State 3S_1 <i>triplet</i>
$\frac{1}{\sqrt{2}}(1s(r_1)2s(r_2) - 1s(r_2)2s(r_1))\beta_1\beta_2$	$S=1, M_s=-1$	
$\frac{1}{2}(1s(r_1)2s(r_2) - 1s(r_2)2s(r_1))(\alpha_1\beta_2 + \alpha_2\beta_1)$	$S=1, M_s=0$	
$\frac{1}{2}(1s(r_1)2s(r_2) + 1s(r_2)2s(r_1))(\alpha_1\beta_2 - \alpha_2\beta_1)$	$S=0, M_s=0$	State 2^1S_0 <i>singlet</i>

- All states of the spin triplet are degenerate
- It is instructive to calculate the energy difference between the 3S_1 and 2^1S_0 .
- It is necessary to evaluate the expectation value of the Hamiltonian for each state

Helium atom: excited state $1s2s$

- The energy of a given state is just the expectation value of the Hamiltonian:

$$\langle E \rangle = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} = \frac{\left\langle \Psi \left| \left(-\frac{1}{2} \nabla_1^2 - \frac{2}{r_1} \right) + \left(-\frac{1}{2} \nabla_2^2 - \frac{2}{r_2} \right) + \frac{1}{r_{12}} \right| \Psi \right\rangle}{\langle \Psi | \Psi \rangle}$$

- By using the hydrogen $1s$ and $2s$ atomic orbitals as approximate spatial eigenfunctions for He, we arrive at the following energies (after quite a lot of manipulation):

$$\begin{aligned} E(^3S_1) &= E_{1s} + E_{2s} + J - K \\ E(^1S_0) &= E_{1s} + E_{2s} + J + K \\ \Delta E &= E(^3S_1) - E(^1S_0) = -2K \end{aligned}$$

Where:

J = Coloumb integral
 K = Exchange integral

Helium atom: excited state $1s2s$

where E_{1s}, E_{2s} are just the energies of the one-electron $1s$ and $2s$ orbitals, J is the Coulomb integral:

$$J = \int 1s(r_1)^* 2s(r_2)^* \left(\frac{1}{r_{12}} \right) 1s(r_1) 2s(r_2) d\tau$$

And K is the Exchange integral:

$$K = \int 1s(r_1)^* 2s(r_2)^* \left(\frac{1}{r_{12}} \right) 1s(r_2) 2s(r_1) d\tau$$

- Thus, in this approximation, the 2^1S_0 state has an energy $2K$ higher than the 3S_1 state
- The energy of the approximate ground state helium atom wave function $\Psi(^1S_0)$ described earlier is -2.75 Ha.

Slater Determinants

- In the early 1930s, Slater introduced the use of determinants to construct anti-symmetric wave functions
- If we have occupied orbitals $\phi_a \phi_b \phi_c, \dots$ then the N-electron wave function is:

$$\Psi(1,2,3, \dots) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_a(1) & \phi_b(1) & \cdots & \phi_N(1) \\ \phi_a(2) & \phi_b(2) & \cdots & \phi_N(2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_a(N) & \phi_b(N) & \cdots & \phi_N(N) \end{vmatrix}$$

- Determinants naturally satisfy the Pauli anti-symmetry principle by two properties

Slater Determinants

- First, the sign of the determinant changes when we interchange any two rows or any two columns
- Second, the determinant is equal to zero if any two rows or any two columns are identical (or are trivially related, e.g. multiplied by a scalar)
- The ground state helium wave function is thus:

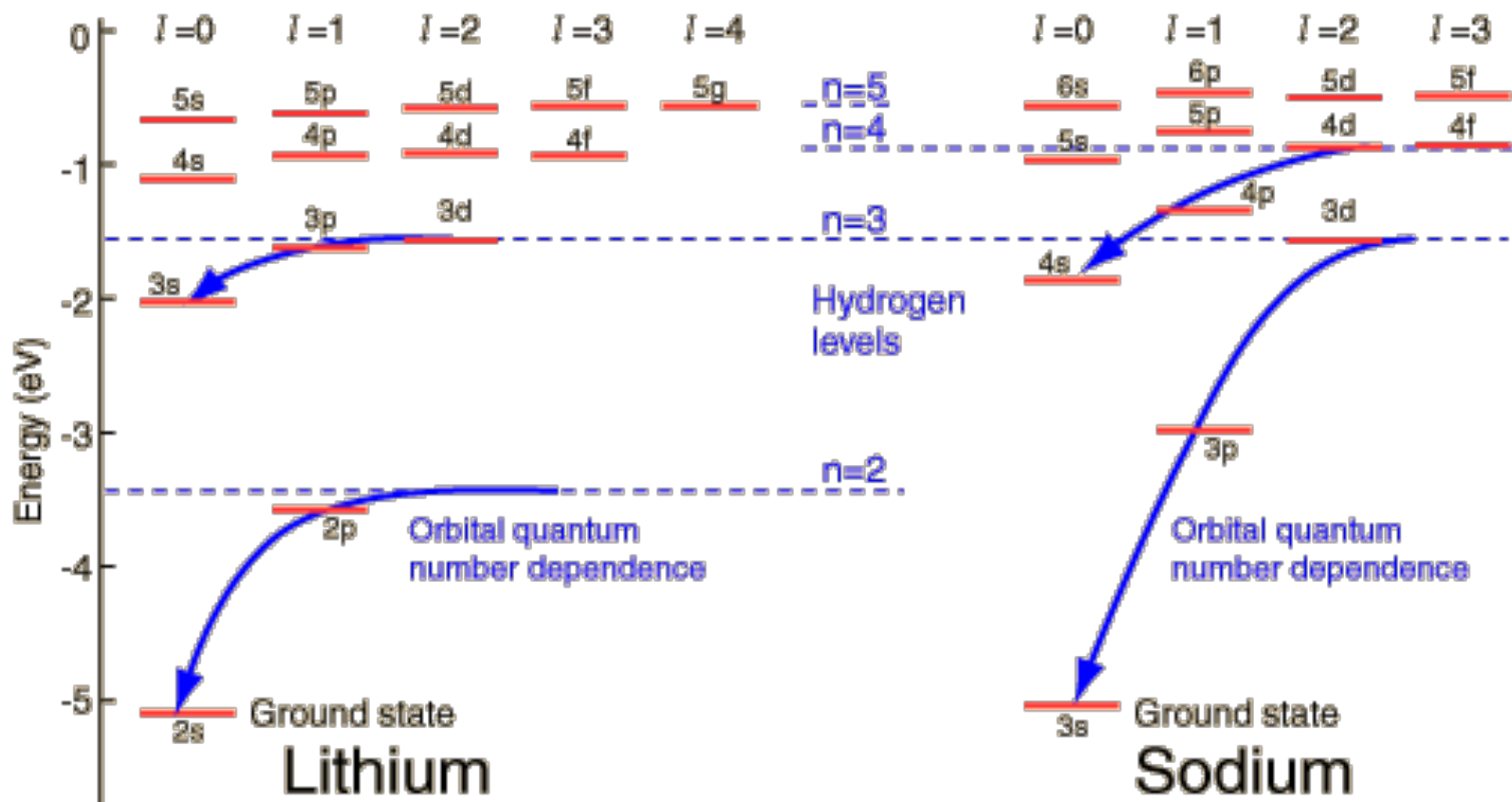
$$\Psi(^1S_0) = \frac{1}{\sqrt{2}} \begin{vmatrix} 1s(r_1)\alpha_1 & 1s(r_1)\beta_1 \\ 1s(r_2)\alpha_2 & 1s(r_2)\beta_2 \end{vmatrix}$$

- The excited singlet state is given by a linear combination:

$$\Psi(2^1S_0) = \frac{1}{2} \left(\begin{vmatrix} 1s(r_1)\alpha_1 & 2s(r_1)\beta_1 \\ 1s(r_2)\alpha_2 & 2s(r_2)\beta_2 \end{vmatrix} - \begin{vmatrix} 1s(r_1)\beta_1 & 2s(r_1)\alpha_1 \\ 1s(r_2)\beta_2 & 2s(r_2)\alpha_2 \end{vmatrix} \right)$$

Orbital dependence of energy

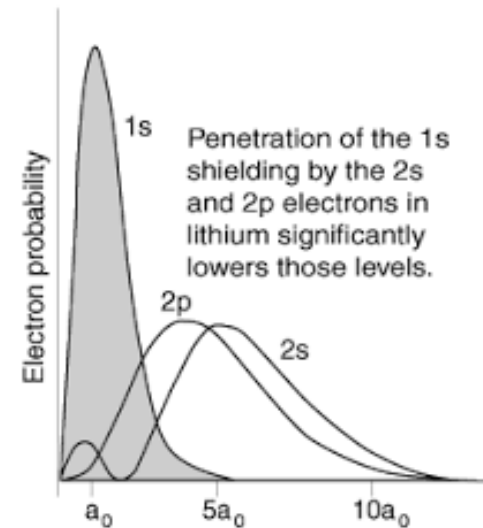
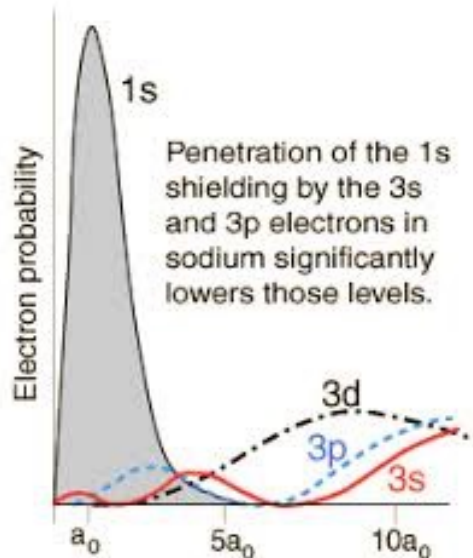
- Alkali metals have a single electron in the outer shell and so we might expect their energy levels to resemble those of hydrogen



Alkali Metals

- Instead, the electron-electron repulsion term has broken the spherical symmetry and we see a strong *orbital dependence of the energy*. (This effect is not present in one-electron atoms, where the energy depends only on the main quantum number n for all $2l + 1$ orbital states)
- For a given n , $E_{ns} < E_{np} < E_{nd} < E_{nf} \dots$, because the penetration of the outer electrons to the nucleus decreases in the order:

$$s > p > d > f > \dots$$



Hartree-Fock approximation for multi-electronic systems:

- We have seen that the ground state wave function of a multi-electronic system can be expressed in terms of a Slater determinant. That is if we have occupied molecular orbitals $\phi_a, \phi_b, \phi_c, \dots$, then the N electron wave function is:

$$\Psi(1,2,3 \dots N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_a(1) & \phi_b(1) & \cdots & \phi_N(1) \\ \phi_a(2) & \phi_b(2) & \cdots & \phi_N(2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_a(N) & \phi_b(N) & \cdots & \phi_N(N) \end{vmatrix}$$

- While the determinant automatically satisfies the Pauli exclusion principle, it does not account for *electron correlation*

Hartree-Fock approximation for multi-electronic systems:

- The Slater determinant is normalized:

$$\langle \Psi(1,2,3 \dots N) | \Psi(1,2,3 \dots N) \rangle = 1$$

- Note that the total Hamiltonian for a multi-electronic system (in atomic units) is:

$$\hat{H} = - \sum_{\alpha}^{N_{nuc}} \nabla_{\alpha}^2 + \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|} - \sum_i^{N_{elec}} \nabla_i^2 - \sum_{\alpha}^{N_{nuc}} \sum_i^{N_{elec}} \frac{Z_{\alpha}}{|r_i - r_{\alpha}|} + \frac{1}{2} \sum_i^{N_{elec}} \sum_{j \neq i}^{N_{elec}} \frac{1}{|r_i - r_j|}$$

Born-Oppenheimer approximation

- Electrons are much lighter than nuclei (the nucleus of H is about 2000 times heavier than the electron). Therefore electrons move much faster than nuclei.
- We consider that nuclei are static, i.e. do not move at all.
- The electrons are considered to adjust immediately to any new configuration of the nuclei.
- Within the BO approximation, the following terms are:

$$\sum_{\alpha}^{N_{nuc}} \nabla_{\alpha}^2 = 0; \quad \text{and} \quad \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|} = \text{constant}$$

Born-Oppenheimer approximation

- The resulting Hamiltonian is:

$$\begin{aligned}
 \hat{H} &= - \sum_i^{N_{elec}} \nabla_i^2 - \sum_{\alpha}^{N_{nuc}} \sum_i^{N_{elec}} \frac{Z_{\alpha}}{|r_i - r_{\alpha}|} + \frac{1}{2} \sum_i^{N_{elec}} \sum_{j \neq i}^{N_{elec}} \frac{1}{|r_i - r_j|} \\
 &+ \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|} = \\
 &= \sum_i^{N_{elec}} \left(-\nabla_i^2 - \sum_{\alpha}^{N_{nuc}} \frac{Z_{\alpha}}{|r_i - r_{\alpha}|} \right) + \frac{1}{2} \sum_i^{N_{elec}} \sum_{j \neq i}^{N_{elec}} \frac{1}{|r_i - r_j|} + \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|} \\
 &= \sum_i^{N_{elec}} \hat{h}_i + \frac{1}{2} \sum_i^{N_{elec}} \sum_{j \neq i}^{N_{elec}} \hat{g}_{ij} + \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|}
 \end{aligned}$$

Hartree-Fock energy

- The following notations were introduced:

$$\hat{h}_i = -\nabla_i^2 - \sum_{\alpha}^{N_{nuc}} \frac{Z_{\alpha}}{|r_i - r_{\alpha}|}$$

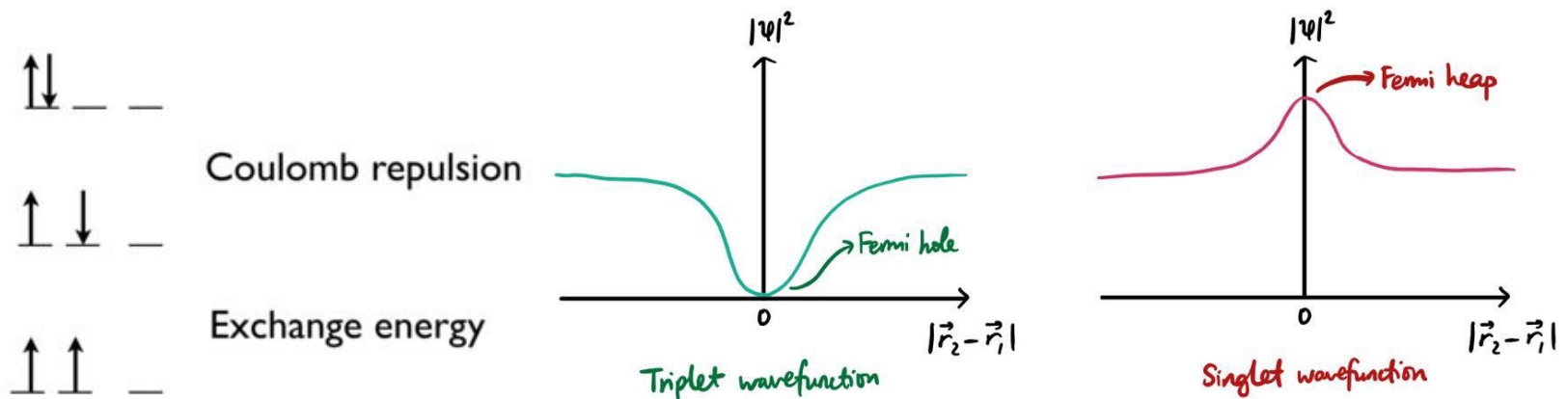
$$\hat{g}_{ij} = \frac{1}{|r_i - r_j|}$$

We proceed now to evaluate the expectation value of the Hamiltonian (i.e. the total energy):

$$E = \langle \Psi(1,2,3 \dots N) | \hat{H} | \Psi(1,2,3 \dots N) \rangle$$

Hartree-Fock approximation

- Electrons move in an average electrostatic field created by all nuclei and remaining electrons
- Each electron does not feel explicitly the presence of other electrons, nor knows it where the other electrons are
- In reality, it is a fact that the probability of finding two electrons close to each other is very small



Hartree-Fock energy

$$\begin{aligned} E &= \langle \Psi(1,2,3 \dots N) | \hat{H} | \Psi(1,2,3 \dots N) \rangle \\ &= \left\langle \Psi(1,2,3 \dots N) \left| \sum_i^{N_{elec}} \hat{h}_i + \frac{1}{2} \sum_i^{N_{elec}} \sum_{j \neq i}^{N_{elec}} \hat{g}_{ij} + \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|} \right| \Psi(1,2,3 \dots N) \right\rangle \\ &= \sum_i^{N_{elec}} \langle \Psi(1,2,3 \dots N) | \hat{h}_i | \Psi(1,2,3 \dots N) \rangle + \\ &\quad + \frac{1}{2} \sum_i^{N_{elec}} \sum_{j \neq i}^{N_{elec}} \langle \Psi(1,2,3 \dots N) | \hat{g}_{ij} | \Psi(1,2,3 \dots N) \rangle + \\ &\quad + \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|} \end{aligned}$$

Hartree-Fock energy

- After some mathematical manipulation, the previous equation is simplified to (for closed shell molecules):

$$E = 2 \sum_i^{N_{elec}/2} H_i + \sum_i^{N_{elec}/2} \sum_j^{N_{elec}/2} (2J_{ij} - K_{ij}) + V_{NN}$$

Where:

$$H_i = \langle \phi_i(1) | \hat{h}_i | \phi_i(1) \rangle \quad = \text{core Hamiltonian} = \text{monoelectronic integrals}$$

$$\left. \begin{aligned} J_{ij} &= \langle \phi_i(1)\phi_j(2) | \hat{g}_{ij} | \phi_i(1)\phi_j(2) \rangle \\ K_{ij} &= \langle \phi_i(1)\phi_j(2) | \hat{g}_{ij} | \phi_j(1)\phi_i(2) \rangle \end{aligned} \right\} = \text{Electron Repulsion Integrals} = \text{bielectronic integrals}$$

$$V_{NN} = \frac{1}{2} \sum_{\alpha}^{N_{nuc}} \sum_{\beta \neq \alpha}^{N_{nuc}} \frac{Z_{\alpha} Z_{\beta}}{|r_{\alpha} - r_{\beta}|}$$

Hartree-Fock energy

- Note that ϕ_i is a molecular orbital whose form is yet to be determined. Determination of ϕ_i requires minimization of the total energy with respect to molecular orbital coefficients, following the variational principle. This leads to the following equation:

$$\hat{F}\phi_i(1) = \varepsilon_i\phi_i(1)$$

where ε_i is the orbital energy and $\hat{F}$ is called Fock operator, which can be expressed as:

$$\hat{F} = -\nabla_1^2 - \sum_{\alpha}^{N_{nuc}} \frac{Z_{\alpha}}{|r_1 - r_{\alpha}|} + \sum_j^{N_{elec}/2} (2J_j(1) - K_j(1))$$

Hartree-Fock energy

$$\hat{F} = -\nabla_1^2 - \sum_{\alpha}^{N_{nuc}} \frac{Z_{\alpha}}{|r_1 - r_{\alpha}|} + \sum_j^{N_{elec}/2} (2J_j(1) - K_j(1))$$

$$J_j(1)f(1) = f(1) \int \frac{|f_j(2)|^2}{|r_1 - r_2|} dr_2$$

$$K_j(1)f(1) = f_j(1) \int \frac{f_j^*(2)f(2)}{|r_1 - r_2|} dr_2$$

Total energy is a sum of individual orbital energies:

$$E = \sum_i^{N_{occ}} \gamma_i \varepsilon_i ,$$

Where γ_i –occupancy of the orbital i

Hartree-Fock-Roothaan Equation

Hartree-Fock equation is difficult to solve. In 1955, Roothaan proposed to use a linear combination of atomic orbitals (LCAO) to represent the molecular orbitals:

$$\phi_i = \sum_s^N c_{si} \chi_s$$

Substituting this form of the MO into the Hartree-Fock equations:

$$\sum_s^N c_{si} \hat{F} \chi_s = \varepsilon_i \sum_s^N c_{si} \chi_s$$

Multiplication by χ_t^* and integration yields:

$$\sum_s^N c_{si} (F_{ts} - \varepsilon_i S_{ts}) = 0, \quad t = 1, 2, 3, \dots, N$$

Hartree-Fock-Roothaan Equation

$$\sum_s^N c_{si}(F_{ts} - \varepsilon_i S_{ts}) = 0, \quad t = 1, 2, 3, \dots, N$$

Where:

$$F_{ts} = \langle \chi_t | \hat{F} | \chi_s \rangle$$

$$S_{ts} = \langle \chi_t | \chi_s \rangle$$

For HFR Equation to have a non trivial solution, the secular determinant must be zero:

$$|F_{ts} - \varepsilon_i S_{ts}| = 0$$

Highlighted equations in the last several slides *can be solved iteratively*, using a self-consistent field (SCF) procedure to make the total energy convergent to a tolerant value. These equations form the basis of modern quantum & computational chemistry.

Hartree-Fock-Roothaan Equation

- From HFR equation, we are required to evaluate the integrals J and K and to obtain the coefficients c_{si} ($s, i = 1, 2, 3, \dots, N$)

$$\hat{F} = -\nabla_1^2 - \sum_{\alpha}^{N_{nuc}} \frac{Z_{\alpha}}{|r_1 - r_{\alpha}|} + \sum_j^{N_{elec}/2} (2J_j(1) - K_j(1))$$

$$J_j(1)f(1) = f(1) \int \frac{|f_j(2)|^2}{|r_1 - r_2|} dr_2$$

$$K_j(1)f(1) = f_j(1) \int \frac{f_j^*(2)f(2)}{|r_1 - r_2|} dr_2$$

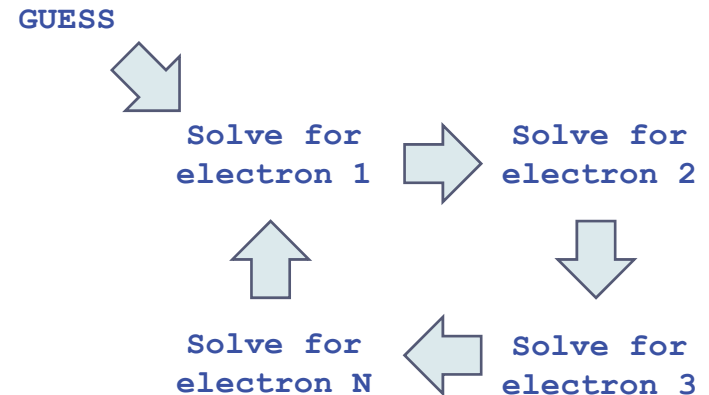
$$\phi_i = \sum_s^N c_{si} \chi_s$$

- Calculations of these equations are still quite difficult to solve, because a large number of these integrals must be computed. More approximations are devised.

Hartree-Fock-Roothaan Equation

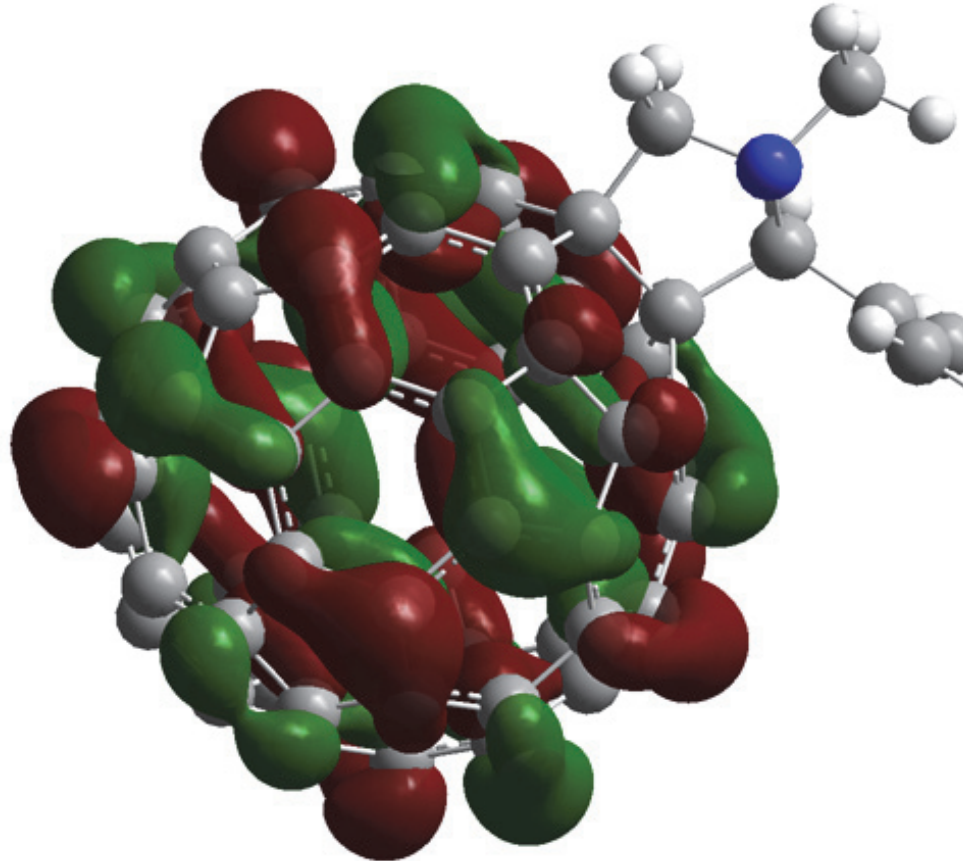
Solving this *for a single electron in an average electric field of all other electrons* is relatively easy, and HF method proceeds by treating each electron in turn in a cycle (see the schematic on the right).

The algorithm updates the wavefunctions for each electron until the solution no longer changes. This procedure is known as the self-consistent field method.



Hartree-Fock-Roothaan Equation

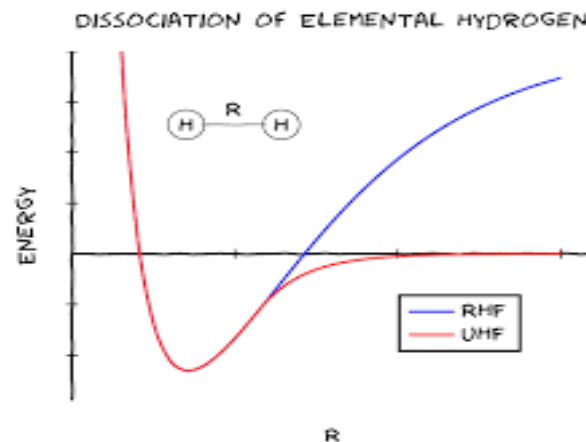
One of the Hartree-Fock orbitals in a fullerene derivative.



Spin Restricted versus Spin Unrestricted

Spin-restricted: each occupied molecular orbital has two electrons. Spin-alpha density is exactly identical to spin-beta density. The shape of spin-alpha orbitals is identical to spin-beta orbitals.

Spin-unrestricted: Two sets of independent molecular orbitals are optimized. Spin-alpha and spin-beta orbitals. In large, at electronic ground state and equilibrium geometry, RHF and UHF are equivalent. However, UHF performs much better near dissociated limit. The shape of spin-beta orbitals is identical to spin-alpha orbitals. UHF wave function is subject to spin-contamination problem.



Practical Session

- A few words about Linux OS: folders, executables, variables, profile, etc.
- Installation of the ORCA / MOLCAS software on NSCC.
- Preparation of a simple calculation. Description of the input. Description of the job submission script.
- Submitting a simple test calculation. PBS job scheduler.
- Description of the output.

A few words about Linux OS

- Derives from Unix. Main developer -- Linus Torvald – started this project for himself, but later he made it open.
- Many flavors of Linux exist, but the “Linux Kernel” (which contains core functions) is the same.

Positive moments:

- It is free and open source. Many software are also free and open-source.
- Command line (i.e. the Terminal) allows direct interaction with the hardware: create a file, execute a program, kill a certain process, etc.

Drawbacks:

- PC Games and some professional software do not have Linux equivalents: MS Office, Adobe, Corel, EndNote, OriginLab, etc. Equivalent free programs might lack some functionality compared to the professional packages.
- Users need to learn various things before they become productive.

Linux folder convention:

Upon login the user arrives in `$HOME` (or `~/`). The user has FULL power to do whatever he pleases inside its own `$HOME`: create files, compile programs , run calculations, etc. Each user of the same server has its own home folder.

`/home/<username>` or `~/` or `$HOME`

Main root folder `“/”` – is the “bottom” folder. Only the superuser (i.e. the root) is able to modify its content. `$HOME` is a subfolder of the “root”. The superuser can do whatever he pleases inside the `“/”` folder and this is dangerous.

`/boot/` -- contains information about booting (starting)

`/usr/` -- contains various compiled and installed software by package managers.

`/bin/` -- contains basic utilities needed by the OS itself.

`/etc/` -- contains usual customisations and local databases

`/opt/` -- places to put local compiled software, so that one can keep track of it.

...

Linux Variables:

`$HOME`

`$PATH` – a list of folders separated by “:” which contains executables. Only binary files which are included in the user’s `PATH` are found by OS and thus could be executed.

How to add a folder to your `PATH`:

```
export PATH=<desired folder>:$PATH
```

Example:

```
echo $PATH
```

```
PATH:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

Add ORCA to your `PATH`:

```
export ORCA=$HOME/software/orca_5_0_3_linux_x86-64_openmpi411
```

```
export PATH=$ORCA/:$PATH
```

```
echo $PATH
```

Linux Variables:

- Check the presence of a certain program or utility:

`which orca`

`ldd $ORCA/orca`

If some libraries of ORCA were not found, then the execution of ORCA will crash. Nothing is wrong here. The OS simply does not know where to look for the required libraries.

`LD_LIBRARY_PATH` is telling the operating system which additional folders are eligible to be searched for libraries during the code's execution.

`echo $LD_LIBRARY_PATH`

`export LD_LIBRARY_PATH=<desired folder>:$LD_LIBRARY_PATH`

Check again:

`ldd $ORCA/orca`

User's experience customization:

`$HOME/.bash_profile`

(user-defined commands, aliases, PATH and LD_LIBRARY_PATH, etc.)

`$HOME/.bashrc`

(non-default values of some program's variables)

`$HOME/.vimrc`

Many programs contain various customization is hidden files or folders with common name ending with "rc".

DO NOT DELETE these files. If removed, the user might have various issues in his work: programs not found, modules not initialized, etc.)

User's Customisation:

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Many programs contain various customization is hidden files or folders with common name ending with "rc".

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Unix text editor: *vi*

- Most supercomputers have very ascetic installations of *Linux* – more often than not, rich editors, such as ***emacs***, would not be available. All *Unix* systems do, however, have a bootstrap editor called ***vi***...

vi <filename> --- opens the <filename> using the *vi* editor

:q -- closes the file (no changes are saved)

:ZZ -- closes and saves the changes

Shift-I -- starts the “insert mode” (so that you can insert/delete text)

Shift-V -- selects a line

Shift-Y -- copies the selected text in active memory and can “paste it” in another place

...

A ***vi*** cheat sheet is provided in email.

Ask a friendly *Unix* guru to show you how to use ***vi*** – this would save you a considerable amount of time and trouble in all your future dealings with *Unix* systems. Customisation helps a lot.

Unix basic commands

cp <file 1> <file 2>	=> copy <file 1> in <file 2>
mv <file 1> <file 2>	=> move <file 1> in <file 2>
rm <file 1>	=> remove <file 1>
cd <folder 1>	=> change directory to <folder 1>
cd ../	=> move one folder toward the root
chmod +x <file 1>	=> change permission of <file 1> and make it executable
touch <file 1>	=> update the time <file 1> as it was last saved.
zip archive.zip <file 1>	=> add <file 1> to the archive.zip
sh script.sh	=> execute “script.sh” (= ./script.sh)
qsub script.sh	=> submit the script.sh to the queuing system (as a calculation).
qdel JOBID	=> delete the JOBID from the list of jobs.

A list of most common Linux commands are given as a cheatsheet.

Task

- 1) Study the **potential energy surface** of H_2 at RHF level of theory in the domain 0.5-10 Angstrom, at minimum 20 points.

Automate the calculation by the following **bash loop (for submission of gaussian jobs on HPC)**:

```
input=Gaussian_PES_H2_UB3LYP

for R in 0.50 0.55 0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95 1.00 1.05 1.10 1.15 1.20 1.25 ; do
    sed -e "s/DISTANCE/$R/g" $input.gjf > "$input"_R_"$R".gjf
    g16 < "$input"_R_"$R".gjf > "$input"_R_"$R".log
    echo "The calculation is done!"
done
grep 'SCF Done' "$input"*.log > results.txt
```

Input files for Gaussian PES/ H_2 are uploaded in Canvas.

- 1) Compare the results with UHF results.
- 2) Keep the same identical basis sets for both cases. Can you explain why the results differ?

As alternative, you may use the function **scan** in Gaussian / GaussView.

Result

